Task 3 EEG forecasting and nominal charge

Does nominal stimulation charge improve prediction of the next EEG voltage mean beyond what the recent EEG already predicts?

Finding. Charge reduced post-trigger squared error by 0.21% to 0.62%. All three participant bootstrap intervals included zero.

Model

For each participant, electrode, and target window, linear regression pooled trials across eligible waveform types. The predictors were three consecutive 50 ms voltage means at the same electrode, immediately before the target.

Target mean = intercept + weighted EEG history + charge slope × Q

The history-only baseline omitted the charge term. Q was nominal cathodic charge, calculated from the recorded peak current, 0.4 ms phase duration and waveform area factor. EEG was scalp-average referenced without trial-baseline subtraction.

Held-out test

1,382 trials from 10 participants and 62 scalp electrodes. Each participant-waveform series was split into five chronological chunks. Every training fold included all three amplitude tiers, and both models used the same test trials. Targets began at 200 or 300 ms after the trigger. Matched pre-trigger targets began at −350 or −250 ms.

Results

Target	History R^2	History + Q R^2	Post error reduction (95% interval)	Pre error reduction
50 ms	0.999699	0.999699	**+0.21%** (-1.03% to +1.62%)	-0.97%
100 ms	0.999532	0.999532	**+0.29%** (-0.92% to +1.79%)	-0.78%
200 ms	0.999034	0.999092	**+0.62%** (-1.04% to +2.81%)	-0.98%

Error reduction compares held-out mean squared error against history alone. Intervals use 10,000 participant bootstrap draws.

Interpretation

Adjacent EEG windows were already highly predictable from voltage history. The charge gains were small and uncertain. This test held out trials within recorded waveform types and leaves transfer to Task 4 untested.